

Math 520, F15

Quiz 10

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find the Laplace transforms of the given functions:

$$(a) f(t) = \begin{cases} 1, & 0 \leq t < \pi \\ 0, & \pi \leq t < \infty \end{cases}$$

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \int_0^{\infty} e^{-st} f(t) dt \\ &= \int_0^{\pi} e^{-st} dt \\ &= \left[-\frac{e^{-st}}{s} \right]_{t=0}^{t=\pi} \\ &= -\frac{e^{-\pi s}}{s} + \frac{1}{s} \\ &= \frac{1 - e^{-\pi s}}{s}, \quad s > 0 \end{aligned}$$

$$(b) f(t) = e^{2t} \cos 3t$$

$$\begin{aligned} \mathcal{L}\{\cos 3t\} &= \frac{s}{s^2 + 3^2} = \frac{s}{s^2 + 9} \\ \Rightarrow \mathcal{L}\{e^{2t} \cos 3t\} &= \frac{s-2}{(s-2)^2 + 9}, \quad s > 2 \end{aligned}$$

Problem 2. (5 points) Find the inverse Laplace transforms of the given functions:

$$(a) F(s) = \frac{6}{(s-1)^3}$$

$$\begin{aligned} \frac{6}{(s-1)^3} &= \frac{2 \cdot 3}{(s-1)^3} \\ \mathcal{L}^{-1}\left\{\frac{2}{s^3}\right\} &= t^2 \\ \Rightarrow \mathcal{L}^{-1}\left\{\frac{6}{(s-1)^3}\right\} &= 3 \mathcal{L}^{-1}\left\{\frac{2}{(s-1)^3}\right\} \\ &= 3e^t t^2 \end{aligned}$$

$$(b) F(s) = \frac{s-2}{s^2 + 3s - 4}$$

$$\begin{aligned} \frac{s-2}{s^2 + 3s - 4} &= \frac{s-2}{(s+4)(s-1)} = \frac{A}{s+4} + \frac{B}{s-1} \\ &= \frac{A(s-1) + B(s+4)}{(s+4)(s-1)} = \frac{(A+B)s + (-A+4B)}{(s+4)(s-1)} \\ \Rightarrow \begin{cases} A+B=1 \\ -A+4B=-2 \end{cases} &\Rightarrow \begin{cases} A=\frac{6}{5} \\ B=-\frac{1}{5} \end{cases} \\ \Rightarrow \mathcal{L}^{-1}\left\{\frac{s-2}{s^2 + 3s - 4}\right\} &= \mathcal{L}^{-1}\left\{\frac{6}{5} \cdot \frac{1}{s+4} - \frac{1}{5} \cdot \frac{1}{s-1}\right\} \\ &= \frac{6}{5} \mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\} - \frac{1}{5} \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} \\ &= \frac{6}{5} e^{-4t} - \frac{1}{5} e^t \end{aligned}$$